Exotic Manifolds and the Parameter Space of Machine Learning

What if the parameter space were $n = 4$ - or 7-dimensional?

A research note for differential geometers and AI research scientists

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Abstract

The parameter space of a deep neural network is almost always treated as $\mathbb{R}^N$ with N ≫ 1, and optimization proceeds by backpropagation together with some descendant of gradient descent. In this note we pursue a thought experiment: suppose the parameter space were homeomorphic to $\mathbb{R}^4$ or to S^7 — dimensions in which a topological manifold admits more than one smooth structure in the sense of differential topology. On which smooth structure is gradient-based optimization actually being performed, and does the choice of structure alter the optima that training converges to?

We argue the following. (i) SGD, Adam, and the natural gradient method all presuppose a single smooth chart on which the update rule is written, so the learner has implicitly committed to one diffeomorphism class before training begins. (ii) In principle the critical-point structure of a loss function depends on the smooth structure carried by its domain, so two smoothings of the same topological space could in principle disagree on minima. (iii) In practice, floating-point arithmetic, bounded architectures, and standard computational graphs conspire to force the standard smooth structure, so exotic structures are not expected to surface as observable phenomena in current systems.

We also clarify that the invariance of Amari's natural gradient — invariance under reparametrization — is a statement about charts within one smooth structure, and is a strictly weaker notion than invariance under change of smooth structure.

Chapter 2 develops the Casson-handle construction behind exotic $\mathbb{R}^4$ in some detail. Appendices A and B record technical corrections aimed at specialists in differential topology and information geometry respectively, each followed by an explanation pitched to AI research scientists.

Scope

This note is a thought experiment. To the author's knowledge, no peer-reviewed paper has directly addressed exotic smooth structures on the parameter space of a machine-learning model. We are accordingly speculative throughout, and we do not claim that the phenomena discussed here have any direct bearing on the day-to-day practice of training deep networks.

Our intended primary reader is an AI research scientist who is comfortable with linear algebra, multivariable calculus, and probability, who has heard of Riemannian metrics and Fisher information matrices, and whose familiarity with topology and manifold theory does not extend much beyond the notions of continuous map and topological space.

The main text (§1 through §8) is written so that such a reader can read it through. Issues of strict mathematical rigor that specialists in differential topology and information geometry would want addressed are isolated in Appendices A and B, where each entry is divided into a precise correction for the specialist and a plain-language commentary for the AI research scientist.

We note also that the central mathematical results we draw on — those of Milnor, Freedman, Donaldson, and Taubes — can be understood at the level needed for our argument without entering into the full Casson-handle construction. The note is organized so that §3 and §4 alone suffice to reach the main claims; the Casson-handle material is collected in §5 as a self-contained optional chapter for the interested reader.

1. Introduction

1.1 The standard setup of gradient-based optimization

The weight vector θ of a deep neural network is an N-dimensional real vector, with N now reaching 10^9 or 10^{12} in the largest contemporary models. We routinely write

$$\theta \in \mathbb{R}^N$$

and consider the optimization problem of minimizing a loss $L : \mathbb{R}^N \to \mathbb{R}$. The notation looks innocuous: it appears to commit us to nothing beyond what is needed to define a vector and a real-valued function. We argue, however, that two distinct mathematical choices have already been made silently by the time we write the symbol $\mathbb{R}^N$, and that the second of these choices is the entry point for the present note.

1.2 An overlooked structural choice

Stochastic gradient descent updates a parameter θ by

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$$

where η is the step size and ∇L is the gradient of L. The right-hand side requires us to interpret ∇L as a vector — a tangent vector, in geometric language — and the very statement that $\nabla L \in$ T_θ $\mathbb{R}^N$ is meaningful presupposes that $\mathbb{R}^N$ has been equipped with a Riemannian metric, since 'gradient' is a metric-dependent notion. The conventional choice is the Euclidean metric ⟨u, v⟩ = Σ u_i v_i, but it is a choice.

Amari's natural gradient method (Amari 1998) recognizes this choice and replaces the Euclidean metric by the Fisher information metric

$$G(\theta) \;=\; \mathbb{E}_{x\sim p(x;\theta)}\big[\nabla_\theta \log p(x;\theta)\, \nabla_\theta \log p(x;\theta)^\top\big]$$

associated to a parametric family p(x; θ). The corresponding update is

$$\theta_{t+1} = \theta_t - \eta\, G(\theta_t)^{-1} \nabla L(\theta_t)$$

Adam, RMSProp, and other adaptive methods can be read as estimating diagonal approximations to a metric on the fly. Whichever method is chosen, the choice is a choice of Riemannian metric on the parameter space.

This is the first structural choice. It has been studied extensively in the information-geometry literature and is well understood by AI research scientists working on optimization.

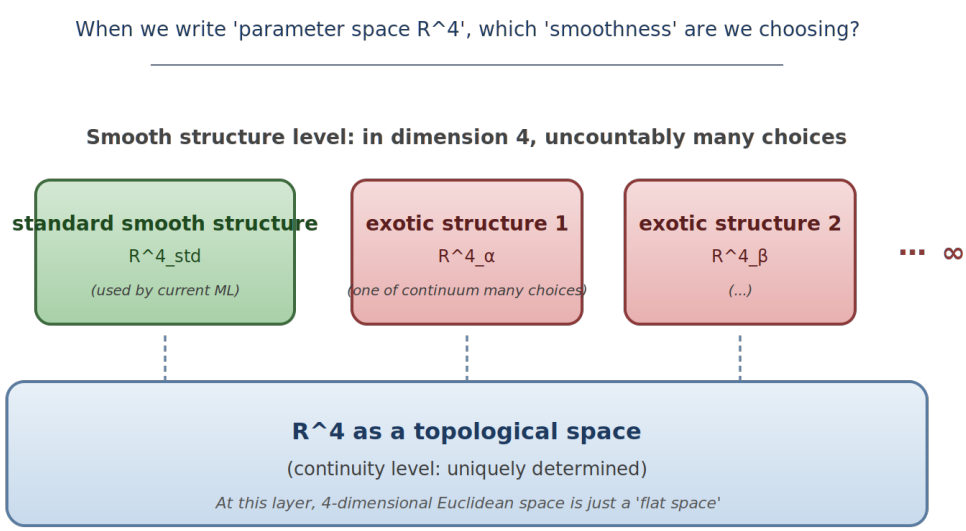
> **◇ What is a Riemannian metric**
>
> A Riemannian metric is a rule that, at each point of a space, prescribes an inner product between two tangent directions. The Euclidean metric ⟨u, v⟩ = Σ u_i v_i is the simplest example: it is the same inner product everywhere. The Fisher metric is more interesting: it assigns to each parameter θ of a probability family p(x; θ) an inner product that depends on θ .
>
> When the metric changes, so do 'distance', 'geodesic', and 'steepest descent direction'. For instance, treating a Gaussian as parametrized by mean μ and standard deviation σ,

> the Euclidean distance from (0, 1) to (1, 1) is exactly 1; the Fisher distance is something else, reflecting how distinguishable the corresponding distributions are.
>
> We will treat the choice of metric as an upper-level structural decision that operates within a fixed smooth structure. The choice of smooth structure itself, discussed in the next subsection, is a deeper, more foundational decision that sits below the metric.

There is, however, a second structural choice lying beneath the choice of metric. Defining a Riemannian metric on a space M presupposes that M has been equipped with a smooth structure — a notion of which functions on M are smooth, of how to differentiate them, and of what the tangent spaces T_x M are. Without a smooth structure, neither 'gradient' nor 'tangent vector' has any meaning.

The conventional choice of smooth structure on $\mathbb{R}^N$ is the standard one inherited from coordinatewise differentiation. We argue that this is also a choice. In dimension 4, in fact, the choice is highly nonunique: the topological space $\mathbb{R}^4$ carries an uncountable family of mutually non-diffeomorphic smooth structures, the so-called exotic $\mathbb{R}^4$'s of Freedman, Donaldson, and Taubes.

Figure 1. Multiple smooth structures on the same topological R^4



Topologically all R^4, yet smoothness (differentiability) admits uncountably many distinct versions.

◇ A first informal picture

From single-variable calculus we are used to the idea that whether a given function is differentiable is a property of the function, not of the line $\mathbb{R}$ on which it lives. The function $f(x) = |x|$ fails to be differentiable at 0; the function $f(x) = x^2$ is smooth everywhere; and these statements appear to be intrinsic to f.

But this depends on having chosen the standard coordinate x on $\mathbb{R}$. If one were to give $\mathbb{R}$ a different coordinate y and judge differentiability with respect to y, the verdict on a given function might change. (For $\mathbb{R}^1$ itself, this freedom is illusory, because $\mathbb{R}^1$ admits, up to diffeomorphism, only one smooth structure.)

Stallings (1962) showed the same for $\mathbb{R}^n$ whenever $n \geq 5$: only one smooth structure, up to diffeomorphism. The same is known for $n \leq 3$ by classical work. So in every dimension other than four, the smooth-structure question has a unique answer and one need not worry about it. Dimension four is the unique exception, and it is exceptional in a strong sense: $\mathbb{R}^4$ admits an uncountable family of distinct smooth structures.

Modern deep learning operates with $N \gg 10^6$, well outside the dimension where this phenomenon occurs, so it does not produce any direct technical issue for current models. But the fact that the most innocuous-looking dimension — four — is the one in which 'smoothness' admits genuine choice is a useful warning about the geometric foundations of the standard setup.

1.3 Three questions

With this picture in place, we organize the rest of the note around three questions.

- **(Q1)** Which smooth structure does the learning algorithm actually run on, and what determines that choice — the algorithm, the implementation, or something else?
- **(Q2)** Can the choice of smooth structure change the optimum θ^* that training converges to?
- **(Q3)** Do different optimization algorithms — SGD, Adam, the natural gradient — differ in their sensitivity to the smooth structure?

The short answers, justified in the body of the note, are: (Q1) the implementation chooses, the algorithm does not; (Q2) yes in principle, but no in any current implementation; (Q3) no, all

current methods are equally insensitive to the smooth structure. Reaching these answers requires some mathematical preparation, which we provide in §2.

1.4 Outline

Section 2 collects the geometric and information-theoretic background needed for the rest of the note, written for readers without prior exposure to differential topology. Section 3 records what is known about the dimensional dependence of smooth structures on Euclidean space, with the unique exceptional behavior at n = 4 stated as the central mathematical fact we will use. Section 4 explains why this fact does not, in current practice, leak into the behavior of gradient-based learning algorithms — the operative word being 'current'. Section 5 is an optional self-contained development of the Casson-handle construction that produces exotic $\mathbb{R}^4$; it can be skipped without loss of continuity. Section 6 returns to the comparison between SGD and the natural gradient, separating two distinct notions of invariance that are sometimes conflated. Section 7 discusses the broader significance of what we have called the invisibility of exotic smooth structures in machine learning. Section 8 concludes.

2. Mathematical Preliminaries

This section collects the differential-geometric background needed for the rest of the note. The treatment is informal, with the goal of conveying enough vocabulary and intuition that an AI research scientist can read §3 through §8 without further reference. Readers familiar with smooth manifolds and the basics of information geometry should skip directly to §3.

2.1 Smooth manifolds and smooth structures

A *topological manifold* is a topological space that locally looks like $\mathbb{R}^n$. More precisely, every point has an open neighborhood homeomorphic to an open subset of $\mathbb{R}^n$, and the dimension n is well-defined and constant on connected components. The familiar examples — $\mathbb{R}^N$ itself, the spheres S^7 and S^4, the orthogonal group SO(n), and the open subset of $\mathbb{R}^N$ on which a parametric family p(x; θ) is well-defined — are all topological manifolds.

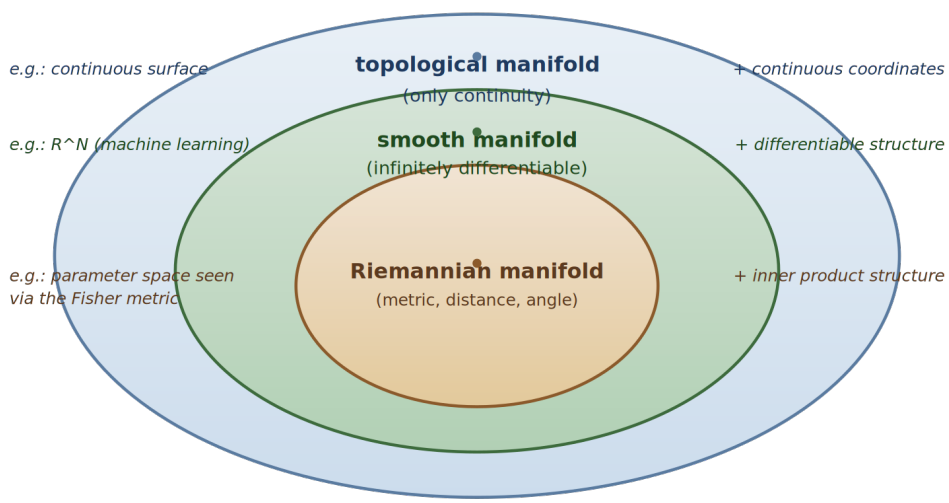
A *smooth structure* on a topological manifold M is a maximal atlas of charts whose transition maps are infinitely differentiable. A topological manifold equipped with a smooth structure is a *smooth manifold*. Once a smooth structure has been fixed, one has unambiguous notions of

smooth function, tangent vector, vector field, and differential form on M.

The conceptual point we wish to emphasize is the layered structure: the topological manifold determines the underlying point set and its continuity properties, while the smooth structure is an additional piece of data that determines what differentiation means. It is entirely possible — though as we shall see, very rare — for one topological manifold to admit several inequivalent smooth structures, and when it does, the choice of structure affects which functions count as differentiable, which maps count as diffeomorphisms, and which submanifolds can be smoothly embedded.

This is the central conceptual point of the present note. Two manifolds may be the same as topological spaces and yet be different as smooth manifolds, and the difference can be invisible to anyone who works only with the underlying point set.

Figure 2. The hierarchy of manifold structures



Each layer adds extra structure to the layer above. The note's main subject is the middle layer (smooth manifolds).

> **◇ Why this matters in machine learning**
>
> It can feel artificial to treat the weight space $\mathbb{R}^N$ of a neural network as a manifold, since it is in the first instance just a vector space. The point of doing so is that gradient-based optimization is naturally a flow on a Riemannian manifold, and quantities such as critical

> points, Hessians, geodesics, and curvature acquire intrinsic meaning only after a smooth and metric structure has been chosen.
>
> Information geometry, originating with Amari (1998) and Amari–Nagaoka (2000), pursues exactly this perspective. The space of parametric distributions {p(x; θ)} is treated as a smooth manifold, equipped with the Fisher metric, and gradient flow with respect to the Fisher metric becomes the natural gradient. All of this presupposes a fixed smooth structure on the parameter space — typically the standard structure on $\mathbb{R}^N$. The present note can be read as an attempt to bring that presupposition into the foreground.

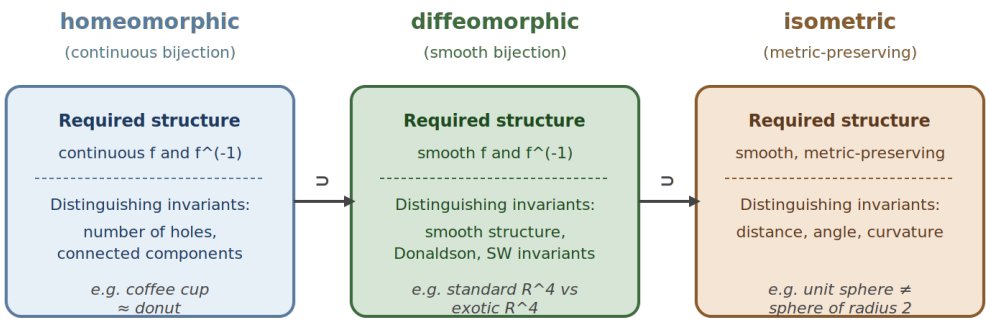
2.2 Equivalences: homeomorphism, diffeomorphism, isometry

Two manifolds may be 'the same' in any of three increasingly demanding senses, corresponding to the three layers of structure introduced above.

- **Homeomorphic**: there is a continuous bijection between them with continuous inverse. The classical caricature is that a coffee cup and a doughnut are homeomorphic. This is the level at which only continuity (and hence topological invariants such as the number of holes or connected components) is preserved.
- **Diffeomorphic**: there is a smooth bijection between them with smooth inverse. This requires both manifolds to be smooth and the comparison map to respect smoothness in both directions. Diffeomorphic smooth manifolds are interchangeable for all purposes of differential geometry.
- **Isometric**: diffeomorphic, and the comparison map preserves a chosen Riemannian metric. This preserves distances, angles, and curvatures.

These are nested: isometric implies diffeomorphic implies homeomorphic, but the converse implications fail in general. The central mathematical fact behind this note is that the failure of 'homeomorphic implies diffeomorphic' has a particularly dramatic form in dimension four: there are uncountably many smooth manifolds that are mutually homeomorphic to $\mathbb{R}^4$ and yet pairwise non-diffeomorphic.

Figure 3. Three levels of "sameness" between manifolds



Conditions strengthen rightward: isometric ⇒ diffeomorphic ⇒ homeomorphic (converses fail)

Central fact of this note: standard R^4 and exotic R^4 are homeomorphic but not diffeomorphic.

◇ The coffee cup and the doughnut

Coffee cups and doughnuts are homeomorphic — a continuous deformation can take one to the other. This is the canonical example of a homeomorphism.

Whether they are diffeomorphic depends on whether the deformation can be carried out smoothly, that is, without introducing creases or sharp corners along the way. If the coffee cup has a smooth surface (no edges where the handle meets the body), then yes. If the cup has a literal sharp seam at the join, then no.

The question of whether they are isometric is more demanding still and almost always negative: the deformation distorts distances on the surface.

◇ Why N = 4 is special (a preview)

In high dimensions (N ≥ 5), Stallings (1962) showed that $\mathbb{R}^N$ admits a unique smooth structure up to diffeomorphism. The same is true in low dimensions (N ≤ 3) by classical work. Only N = 4 is exceptional: $\mathbb{R}^4$ admits an uncountable family of smooth structures.

Why N = 4 is the lone exception is detailed in §3 and §5. Informally, the reason is a delicate dimension count: in dimension 4 'there is exactly no room' for two 2-dimensional objects to be moved past each other, where in higher dimensions there would be enough ambient space and in lower dimensions the configuration cannot arise.

2.3 Riemannian metrics and gradients

A *Riemannian metric* on a smooth manifold M is a smooth assignment, to each point x ∈ M, of an inner product g_x on the tangent space T_x M. A smooth manifold equipped with a Riemannian metric is a *Riemannian manifold*.

Once a Riemannian metric g has been chosen, one can define the *gradient* of a smooth function L : M → ℝ at a point x: it is the unique tangent vector grad_g L(x) ∈ T_x M satisfying g_x(grad_g L(x), v) = (dL)_x(v) for every v ∈ T_x M. The gradient flow

$$\dot{\theta}(t) \;=\; -\operatorname{grad}_g L(\theta(t))$$

is then the continuous-time analogue of gradient descent.

Crucially, the gradient depends on the metric. The same smooth function on the same smooth manifold has different gradients with respect to different Riemannian metrics. SGD with the standard Euclidean metric and the natural gradient with the Fisher metric are two instances of this dependence; their trajectories on the same loss function are in general different.

2.4 The Fisher information metric

Let {p(x; θ) : θ ∈ Θ} be a parametric family of probability densities on a sample space, with Θ an open subset of $\mathbb{R}^N$. The *Fisher information matrix* at θ is

$$G(\theta) \;=\; \mathbb{E}_{x\sim p(x;\theta)}\big[\nabla_\theta \log p(x;\theta)\,\nabla_\theta \log p(x;\theta)^\top\big]$$

where ∇_θ log p(x; θ) is the score function. Treating G(θ) as a Riemannian metric on Θ — that is, taking g_θ(u, v) := u^T G(θ) v on T_θ Θ — gives the parameter space a Riemannian structure that reflects the statistical geometry of the family.

The natural gradient method computes the steepest-descent direction with respect to the Fisher metric:

$$\theta_{t+1} = \theta_t - \eta\, G(\theta_t)^{-1}\,\nabla L(\theta_t)$$

The principal advantage, established by Amari (1998), is invariance under reparametrization of the family — a property we discuss in detail in §6.

2.5 Exotic smooth structures: a first look

We can now state the central object of this note. Two smooth manifolds M and N are said to be *exotic relative to each other* if they are homeomorphic but not diffeomorphic. When the underlying topological manifold is a familiar one such as $\mathbb{R}^4$ or S^7, and one of the smooth structures is the standard one, the other is called an *exotic smooth structure* on that topological manifold.

Milnor (1956) constructed the first exotic smooth structures, on the 7-sphere S^7. Freedman (1982) and Donaldson (1983), in combination, established the existence of exotic smooth structures on the 4-dimensional Euclidean space $\mathbb{R}^4$. Subsequent work by Gompf (1985), Taubes (1987), and DeMichelis–Freedman (1992) showed that there are in fact uncountably many such structures, and that they sit inside the standard $\mathbb{R}^4$ as open subsets.

We turn in §3 to the precise statements of these results and to what they imply about the dimensional dependence of the smooth-structure question, postponing the detailed Casson-handle construction behind them to §5.

3. The Peculiarity of Dimension Four

This section records the central mathematical fact behind the present note: the question of how many smooth structures the topological space $\mathbb{R}^n$ admits has a uniform answer of one in every dimension other than four, and an answer of uncountably many in dimension four. We state the relevant theorems but do not give their proofs; the construction underlying the four-dimensional case is the subject of §5.

3.1 Stability in other dimensions

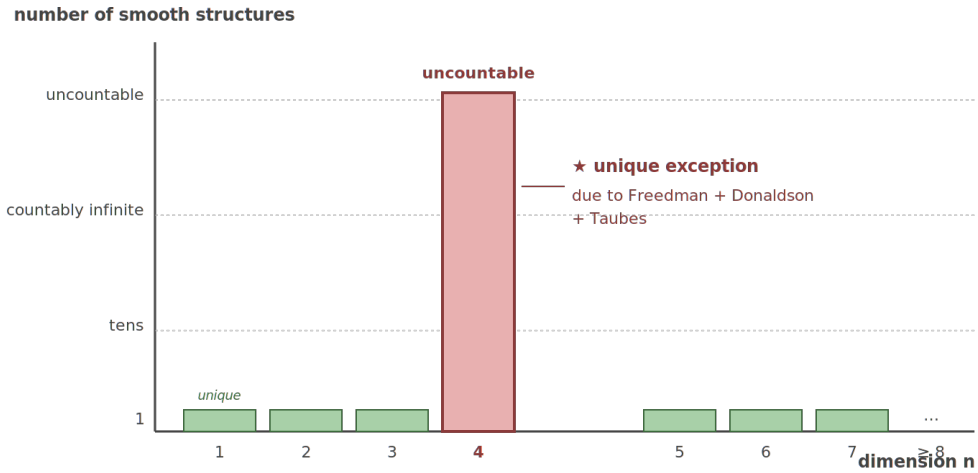
The dependence of the number of smooth structures on $\mathbb{R}^n$ on the dimension n is summarized in the following table.

- n = 1, 2, 3: $\mathbb{R}^n$ admits a unique smooth structure up to diffeomorphism, by classical work of Moise, Smale, and others.
- n ≥ 5: $\mathbb{R}^n$ admits a unique smooth structure up to diffeomorphism, by Stallings (1962).

- n = 4: $\mathbb{R}^4$ admits an *uncountable* family of mutually non-diffeomorphic smooth structures (Freedman 1982; Donaldson 1983; Gompf 1985; Taubes 1987; DeMichelis–Freedman 1992).

Dimension four is therefore the unique outlier.

Figure 4. Number of smooth structures on R^n by dimension n



All other dimensions (1, 2, 3, 5, 6, 7, ...): R^n admits a unique smooth structure (Stallings 1962, etc.)

Dimension 4 alone is the absolute exception, with continuum-cardinality freedom. This is the starting point of the note.

◇ Why dimension four is special

The technical source of this peculiarity is a dimension count. Two 2-dimensional submanifolds A and B of an ambient n-manifold X meet transversely in a 0-dimensional set, and one tries to remove their intersection points by an isotopy that pushes A across a 2-dimensional Whitney disk D bounded by an arc on A and an arc on B.

For the Whitney trick to succeed in the smooth category, D must be embedded in X, and the isotopy must take place in a neighborhood of D that does not meet A or B except along the prescribed boundary arcs. This requires dim D + dim A ≤ dim X and dim D + dim B ≤ dim X — that is, 2 + 2 ≤ n, or n ≥ 5.

When n = 4, the inequality is tight. Generically the disk D meets A and B at additional interior points, and the attempt to remove those new intersections with further Whitney disks produces yet more intersections, so the procedure does not terminate. When n ≤ 3,

> the dimensions are too low for the issue even to arise.
>
> Casson's contribution (1973, published 1986) was to organize this nonterminating bookkeeping into a precise infinite combinatorial object — what we now call a Casson handle — and Freedman's contribution (1982) was to show that the resulting infinite object is, after all, topologically the same as a standard 2-handle. We return to this construction in §5; for §3 and §4 it is enough to know that it exists.

3.2 Milnor's exotic 7-sphere

Historically the first example of an exotic smooth structure on a familiar topological manifold is Milnor's exotic 7-sphere.

> **Milnor (1956).** *There exists a smooth 7-manifold Σ^7 that is homeomorphic to S^7 but not diffeomorphic to it.*

Milnor's construction realizes Σ^7 as the total space of an S^3-bundle over S^3 coming from a quaternionic Hopf fibration; the obstruction to diffeomorphism with the standard S^7 is a Pontryagin class invariant. Subsequent work of Kervaire–Milnor (1963) classified all such structures: smooth structures on S^7 up to orientation-preserving diffeomorphism form a cyclic group of order 28 under connected sum,

$$\Theta_7 \cong \mathbb{Z}/28\mathbb{Z}$$

We mention this here primarily as a finite-dimensional reference point. The 4-dimensional case will be qualitatively different — uncountably many structures rather than 28 — and this distinction is itself a sign of how exceptional dimension four is.

> **◇ What "28 structures" means**
>
> Even mathematicians find the number 28 surprising on first encounter. Why precisely 28 — rather than, say, 17 or 64 — requires K-theory and stable homotopy theory to explain in detail, and we do not pursue it here.
>
> What matters for our purposes is the qualitative content: (i) S^7 admits smooth structures distinct from the standard one, and (ii) the number of such structures is finite,

> indeed exactly 28.
>
> If one were to consider a hypothetical machine-learning model with parameter space S^7 — for instance a model with normalized 8-dimensional embeddings — choosing a Milnor structure rather than the standard one would in principle produce different training trajectories. Real ML systems with N = 7 are essentially nonexistent, so this is interesting only as a thought experiment.

3.3 Freedman, Donaldson, and exotic $\mathbb{R}^4$

We now state the four-dimensional results in the order in which they are typically used in the construction of exotic $\mathbb{R}^4$.

3.3.1 Freedman's classification of topological 4-manifolds

Freedman (1982) gave a complete classification of simply connected closed topological 4-manifolds in terms of their intersection form. The result that matters for us is the following.

> **Freedman (1982).** *For every unimodular symmetric bilinear form Q on a finitely generated free abelian group, there exists a closed simply connected topological 4-manifold X whose intersection form is isomorphic to Q. If Q is even, X is unique up to homeomorphism; if Q is odd, there are exactly two such X up to homeomorphism, distinguished by the Kirby-Siebenmann invariant.*

Freedman's proof works in the topological category, and its central technical input is a procedure for constructing topologically embedded 2-disks where the smooth Whitney trick fails. These topological disks are, as we will see in §5, the topological flattenings of Casson handles.

3.3.2 Donaldson's smooth obstruction

Donaldson (1983), using the Yang–Mills equations of gauge theory, established a strong constraint on smooth closed 4-manifolds that is invisible at the topological level.

> **Donaldson (1983).** *If X is a closed simply connected smooth 4-manifold whose intersection form Q is negative definite, then Q is diagonalizable over the integers,*

$$Q_{E_8 \oplus E_8} \text{ is not realized by any smooth closed oriented 4-manifold}$$

An immediate consequence: the topological 4-manifold whose intersection form is the rank-16 form $E_8 \oplus E_8$ — which is negative definite, even, and not diagonalizable — exists by Freedman's theorem but admits no smooth structure by Donaldson's. The combination of Freedman's existence and Donaldson's obstruction is what produces exotic smooth structures on $\mathbb{R}^4$; the construction excises a suitable open subset of an ambient closed 4-manifold and observes that the inherited smooth structure cannot be diffeomorphic to the standard one.

> **◇ A note on gauge theory**
> Gauge theory originated in particle physics as a framework for describing electromagnetic and Yang–Mills fields. Donaldson adapted it to construct invariants of smooth 4-manifolds by studying the moduli space of self-dual connections on an SU(2) bundle, and showed that the topology of these moduli spaces forces strong constraints on the smooth structure of the underlying 4-manifold.
>
> For our purposes, the relevant input is only the conclusion of Donaldson's theorem and its qualitative consequence: there exist topological 4-manifolds that admit no smooth structure at all. The full machinery of gauge theory is not needed.

3.3.3 Uncountably many exotic $\mathbb{R}^4$'s

Once a single exotic $\mathbb{R}^4$ has been constructed from the Freedman–Donaldson combination, the family can be enlarged. Gompf (1985) showed that the operation of end connected sum produces, from one exotic $\mathbb{R}^4$, a countable family of mutually non-diffeomorphic exotic $\mathbb{R}^4$'s. Taubes (1987) used the moduli of anti-self-dual connections on end-periodic 4-manifolds to show that the family is in fact uncountable. DeMichelis–Freedman (1992) showed that this uncountable family can be realized as open subsets of the standard $\mathbb{R}^4$ itself.

> **Taubes (1987); DeMichelis–Freedman (1992).** *There exists a family {R_α : α ∈ A} of smooth 4-manifolds, each homeomorphic to $\mathbb{R}^4$, such that |A| equals the cardinality of the continuum and the R_α are pairwise non-diffeomorphic. Moreover the R_α can be taken to be open subsets of the standard $\mathbb{R}^4$.*

Informally: the topological space $\mathbb{R}^4$ carries as many distinct notions of 'smooth function' as there are real numbers, and these distinct notions sit inside the standard $\mathbb{R}^4$ as open subsets that are pairwise topologically equivalent but pairwise smoothly inequivalent. By contrast, in every other dimension the family of smooth structures on $\mathbb{R}^n$ is at most finite — in fact a single class for n ≠ 4, and 28 classes for S^7. Dimension four alone exhibits continuum-cardinality freedom; it is the absolute exception.

3.4 Implications for machine learning

We summarize what the preceding results imply for the machine-learning setting.

- Modern deep networks have parameter dimension N in the range 10^6 to 10^{12}, well outside the exceptional dimension n = 4. $\mathbb{R}^N$ in this regime carries a unique smooth structure, so the exotic-structure question does not arise as a practical concern.
- If one were to consider a model with parameter dimension exactly four, in principle one could equip $\mathbb{R}^4$ with any of the uncountably many smooth structures available, and the loss landscape, gradient flow, and critical-point structure would in principle differ across these choices.
- Current implementations — IEEE 754 floating point with automatic differentiation — implicitly fix the standard smooth structure on $\mathbb{R}^N$ before any choice of optimization algorithm is made. The user has no mechanism to select a different structure even if N = 4.
- The reason is structural rather than incidental: the Casson-handle construction underlying exotic $\mathbb{R}^4$ is intrinsically infinite and not expressible by any finite computation graph. We discuss this in detail in the next section.

4. Why Exotic Structures are Invisible to Machine Learning

Section 3 established that $\mathbb{R}^4$ admits uncountably many smooth structures, of which standard machine learning sees only one. This section explains why. The argument has three layers, corresponding to three aspects of the standard implementation: the algorithm, the arithmetic, and the underlying construction of the alternatives.

4.1 SGD and the natural gradient revisited

We begin with the algorithm itself. The SGD update

$$\theta_{t+1} = \theta_t - \eta \nabla L(\theta_t)$$

presupposes the following structural commitments:

- **Linearity.** θ and ∇L are treated as elements of the same vector space, with addition and scalar multiplication well-defined. The parameter space is implemented as a linear space, indeed as $\mathbb{R}^N$.
- **Differentiability.** The symbol ∇L is computed as the vector of partial derivatives $(\partial L/\partial \theta_i)_i$, with each partial derivative implemented as an IEEE 754 floating-point number. For this computation to be meaningful, L must be differentiable as a function on $\mathbb{R}^N$ equipped with its *standard* smooth structure.
- **Inner product.** The product η ∇L and the subtraction from the current θ are both interpreted with respect to the Euclidean inner product. This is the choice of metric — what we called the first structural choice in §1.2.

The first two commitments together fix the standard smooth structure on $\mathbb{R}^N$. This is the second structural choice — and crucially, it is one that no human user explicitly makes. It is built into the ambient computational machinery of IEEE 754 and automatic differentiation.

The natural gradient method does not change this. Its update

$$\theta_{t+1} = \theta_t - \eta\, G(\theta_t)^{-1} \nabla L(\theta_t)$$

replaces the Euclidean inner product with the Fisher metric — that is, it changes the first choice — but the Fisher information matrix G(θ) is still defined as a covariance of partial derivatives in the standard smooth structure of $\mathbb{R}^N$. Adopting the natural gradient changes the first commitment, not the second.

4.2 IEEE 754 and automatic differentiation

Automatic differentiation propagates derivatives through computational graphs by repeated application of the chain rule. It works by associating to each elementary operation (+, ×, exp, log, sin, ...) a known Jacobian, and composing these via known algebraic rules.

For this to be well-defined, the elementary operations must be smooth as functions on $\mathbb{R}^N$ —

and the smoothness here is smoothness with respect to the standard smooth structure. Their Jacobians are computed in standard coordinates.

This presupposition is so deeply built into the technology that it almost looks tautological. The arithmetic statement '3.7 + 1.4 = 5.1' is, of course, addition in the usual sense — there is no discernible alternative.

Our point is that this presupposition is technological, not mathematical. The standard smooth structure of $\mathbb{R}^N$ is built into the IEEE 754 specification, into the C and CUDA implementations of arithmetic, and into the autodiff frameworks built on top of them. None of these layers exposes any handle by which an alternative smooth structure could be specified.

> **◇ What it means for arithmetic to depend on a smooth structure**
> Defining addition on $\mathbb{R}$ requires not only $\mathbb{R}$ as a set but also a topology and a smooth structure on $\mathbb{R}$; the proof that addition is smooth is a statement in the standard smooth structure. If $\mathbb{R}$ were given a different smooth structure (which is a vacuous statement in dimension one but a real one in dimension four), the function $(x, y) \mapsto x + y$ might fail to be smooth, while some other function might become smooth.
>
> Whether IEEE 754 floating-point addition is smooth in an exotic $\mathbb{R}^4$ structure is a meaningful but speculative question. Since no implementation of an exotic smooth structure exists in any computational framework, the question has no concrete answer. It illustrates only that the smoothness we take for granted is smoothness in a particular structure.

4.3 Casson handles and the limits of finite computation

Why is there no implementation of an exotic smooth structure? The answer lies in the construction of exotic $\mathbb{R}^4$ itself. We will examine this construction in detail in §5; here we summarize the structural feature that matters for the present argument.

The construction relies on a Casson handle, a 4-dimensional geometric object built as a countably infinite tower of so-called kinky handles, each level of which is forced by the failure of the Whitney trick at the previous level (see §5 for definitions). The combinatorial data of a Casson handle is a finitely branching infinite signed tree, and the geometric realization is a non-compact

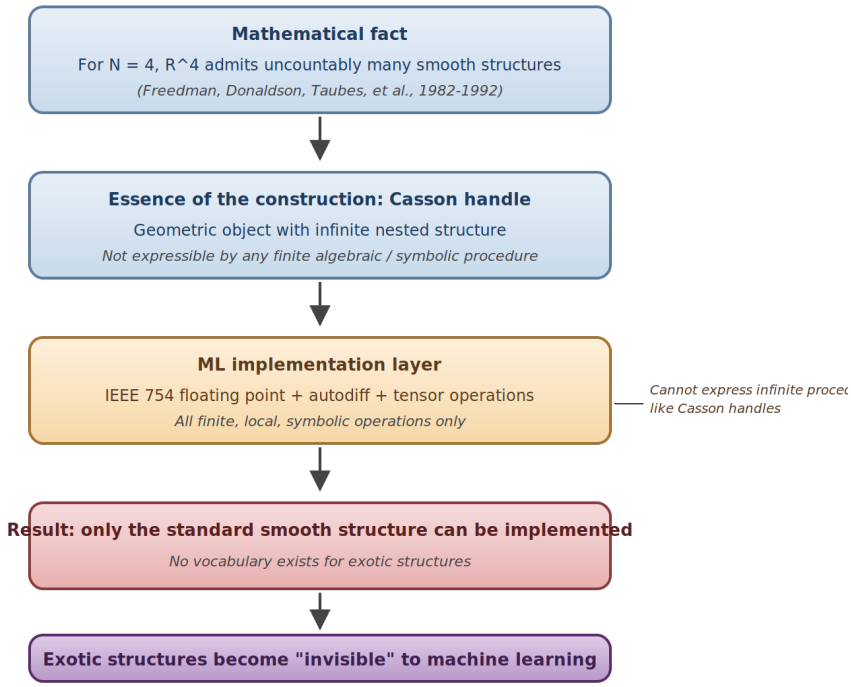
4-manifold whose 'shape' is fundamentally infinite.

The two theorems of Freedman (1982) and Donaldson (1983) that combine to produce exotic $\mathbb{R}^4$ are themselves intrinsically infinite arguments. Freedman's proof is a Bing-style shrinking argument carried out as a limit of countably many small adjustments. Donaldson's proof analyzes the moduli space of solutions to a nonlinear elliptic PDE — the Yang–Mills equation — on a compact 4-manifold and extracts topological information from the geometry of this moduli space. Neither admits a presentation as a finite sequence of algebraic operations on real numbers.

The computational substrate of contemporary machine learning, in contrast, consists entirely of finite, local, symbolic operations on floating-point numbers. Tensor operations, automatic differentiation, optimizer updates — all are finite computational graphs over a finite alphabet of primitives. There is no vocabulary in this substrate for the infinite-tower construction of a Casson handle, and consequently no vocabulary for an exotic $\mathbb{R}^4$ structure.

> **◇ A note on computability**
> We do not mean that Casson handles are uncomputable in the Turing sense. The combinatorial data — a finitely branching signed tree — admits finite encodings, and the tree can be approximated by finite truncations to arbitrary depth.
>
> What we mean is that the geometric object whose existence the construction asserts is not the limit of a sequence of geometric approximations in any sense compatible with finite differentiation. Refining the truncation does not converge to anything that the differentiation primitives in PyTorch or JAX could operate on.
>
> If a future computational paradigm — for instance, one based on synthetic differential geometry, in the spirit of Lawvere and Kock — were to treat smooth structures as first-class objects rather than as implicit consequences of arithmetic primitives, the situation might change. We touch on this prospect in §7.4.

Figure 5. Why machine learning does not "see" exotic structures



4.4 The structural source of invisibility

We now state the central claims of this section, which we will return to in §7.

- **(A)** No contemporary optimization algorithm — SGD, Adam, the natural gradient, or any of their variants — provides a mechanism for selecting a smooth structure on the parameter space. The standard smooth structure is fixed by the implementation stack, not by the algorithm.
- **(B)** In dimension four, the alternatives are uncountable in number and substantively different. Two non-diffeomorphic smoothings of $\mathbb{R}^4$ carry different algebras of smooth functions and therefore in principle different sets of critical points and different gradient flows.
- **(C)** Implementing any alternative would require expressing the Casson-handle construction within the finite vocabulary of contemporary computational frameworks, which is not currently possible and which no existing approach to numerical computation is designed to support.
- **(D)** Therefore exotic smooth structures are, in effect, *invisible* to machine learning as currently practiced. The invisibility is not a deliberate omission but a structural consequence of the implementation choices that define the field.

We turn in §5 to a self-contained development of the Casson-handle construction, for the reader who wishes to see why the alternatives resist finite description. Readers prepared to take this on faith may proceed directly to §6, in which we revisit the comparison between SGD and the natural gradient and clarify the relation between two distinct notions of invariance.

5. The Casson Handle Construction (Optional Deep Dive)

5.1 Reading guide

This section gives a self-contained account of the Casson-handle construction that lies behind exotic $\mathbb{R}^4$. The presentation is mathematical in character, but is pitched at the level of an AI research scientist who has read §2; no prior exposure to differential topology is assumed.

The reader who wishes only to understand the main argument of the note can safely skip this section. Sections 3 and 4 establish the existence and the invisibility of exotic smooth structures without using any feature of the construction beyond its existence. The present section serves three purposes for the reader who continues:

- It exhibits the infinite combinatorial structure that justifies the claim in §4.3 that exotic $\mathbb{R}^4$ resists any finite computational description.
- It shows how the topological flattening theorem of Freedman and the smooth obstruction of Donaldson combine in detail to produce a manifold that is topologically standard but smoothly exotic.
- It introduces vocabulary (kinky handle, Casson tower, h-cobordism) that recurs in the appendices.

Primary sources are Casson's 1973 Cambridge lectures (published as Casson 1986), Freedman (1982), and the textbook treatment in Gompf–Stipsicz (1999).

> **◇ An orientation for the non-specialist**
> The technical density of this chapter is high, so each section ends with a plain-language commentary aimed at readers without a background in differential topology. The intuition to carry through is this: dimension four is oddly cramped. In higher dimensions there is enough room to untangle things you would like to separate, and in lower dimensions (two and three) the spaces are simple enough that the relevant problems never arise. Four is

> the dimension where the things you want to move and the things in the way have exactly matching dimensions, with no room to spare — a precise but unlucky coincidence.
>
> The Casson handle is a device for recording this un-separability as an infinite branching structure. The punchline of the chapter is that this recording makes visible the existence of uncountably many smooth structures on a topological space that is 'obviously just $\mathbb{R}^4$'.

5.2 The failure of the Whitney trick

In dimensions $n \geq 5$, the Whitney trick (Whitney 1944) is the workhorse of smooth manifold theory. Given two submanifolds A and B meeting at two points p, q whose algebraic intersection number is zero, one chooses a 2-dimensional Whitney disk D connecting p and q and isotopes A across D to remove the geometric intersection. Because the ambient dimension is at least five, D has enough room to be used as a guide without being obstructed by A or B.

In dimension $n = 4$, the arithmetic collapses: A, B, and D are all two-dimensional, so D itself generically intersects A and B. The attempt to resolve those new intersections with further Whitney disks generates yet another layer of intersections, and so on. Bookkeeping for this non-termination — organized as a finite, tree-like object — is what Casson introduced, and what the following sections describe.

> **◇ Plain commentary: why four dimensions are special**
>
> Picture two surfaces A and B, each two-dimensional, sitting inside some ambient space and meeting at two points. We want to slide A off B to eliminate those intersections. The standard tool is to find a small two-dimensional disk D whose boundary runs along A and B and use D as a guide for the sliding.
>
> If the ambient space is at least five-dimensional, D has enough room to do its job. With dimensions 2 + 2 = 4 already taken up by A and B, there is at least one dimension left over, which gives D the freedom to be perturbed off any new intersection.
>
> If the ambient space is exactly four-dimensional, there is no room left. D cannot help intersecting A and B at new points, and any attempt to fix those new intersections with further disks creates further intersections. There is no end to the process. Casson's idea

> was to accept this and keep track of the infinite list of failed corrections explicitly.

5.3 Kinky handles

The basic building block in Casson's construction is the kinky handle. We first recall the standard 2-handle for context.

5.3.1 The standard 2-handle

A standard 2-handle is the four-dimensional thickening $D^2 \times D^2$ of a two-dimensional disk, attached to the boundary of a four-manifold along its attaching region $\partial D^2 \times D^2$. Its core is the embedded disk $D^2 \times \{0\}$; its co-core is $\{0\} \times D^2$.

> **◇ Imagining a 2-handle**
> A handle is a building block in topology, in the same sense that a brick is a building block in construction. By stacking handles of different indices (0-handles, 1-handles, 2-handles, and so on) one assembles arbitrary smooth manifolds.
>
> A 4-dimensional 2-handle is, intuitively, 'a 2-dimensional disk thickened in two extra directions to make it 4-dimensional.' The core is the 2-disk at the center; in a standard (non-kinky) 2-handle this core is cleanly embedded — it does not intersect itself.

5.3.2 The kinky handle

A kinky handle K is a 2-handle in which the core has been replaced by an immersion rather than an embedding. More precisely, the core is an immersed 2-disk in the 4-manifold with d transverse self-intersections, where $d \geq 0$ is the kink number. Schematically,

$$K \;\cong\; (D^2 \times D^2) \natural \big(\text{self-plumbings}\big)$$

where $\natural$ denotes boundary connected sum. Each self-intersection is a transverse double point of the core, and contributes one $S^1 \times D^3$ 1-handle to the boundary.

> **◇ Picturing a kinky handle**
> Imagine a hose that you have tried to coil into a flat disk shape, but along the way the hose has passed through itself at two or three places. The flat disk you intended to make has 'kinks' — places where it crosses itself. The kinky handle is the four-dimensional analogue

of this picture.

Why use such a defective object at all? Because in dimension four the Whitney trick fails, and a clean embedded 2-disk is exactly what one cannot find. The kinky handle is the compromise: 'we cannot embed, but we can immerse with finitely many self-intersections, and that is what we will work with.'

An *embedding* is a map that is injective and smooth and whose inverse is smooth on its image. An *immersion* is a map that is locally an embedding but may have global self-intersections. Klein bottles immerse but do not embed in 3-dimensional space; standard knots embed.

5.4 Casson towers

A single kinky handle does not complete the Whitney trick; the d self-intersections remain to be removed. One attempts to remove each of them by an additional Whitney disk, but in dimension four those disks are themselves kinky. Iterating produces an infinite tower.

Formally, a Casson tower is the union of an increasing sequence of stages,

$$\mathrm{CH} = \bigcup_{k=0}^{\infty} K_k, \qquad K_0 \subset K_1 \subset K_2 \subset \cdots$$

where each stage K_{k+1} is the boundary connected sum of K_k with finitely many kinky handles, one attached at each self-intersection of K_k. The combinatorial data of the tower is a finitely branching signed tree: each vertex represents a kinky handle, the number of children of a vertex equals the kink number d of the corresponding handle, and the signs record the intersection signs.

◇ A tempting analogy with deep networks

The layered structure of a Casson tower bears a superficial resemblance to a deep computational graph. Each new stage is built on top of the previous one, with the next layer's structure determined by the failures of the previous one. The analogy is suggestive but breaks down in instructive ways.

- The depth of a deep network is a design choice — one stops at some finite number. The depth of a Casson tower is not chosen by the designer; each stage is forced by the failure of the previous stage, with no stopping rule.

> – The width of a deep network at each layer is fixed by design. The width of a Casson tower at each stage — the kink number d at each vertex — is determined by the previous stage's geometry.
> – A deep network is a finite computational object, evaluable in finite time. A Casson tower is the geometric realization of an inductive process that never terminates.

> It is not an infinite sum that converges; it is the geometric realization of an inductive process that refuses to terminate. This is the concrete content of the claim in §4.3 that a Casson handle does not lift to a finite computation.

5.5 Freedman's topological flattening

The deepest result in this circle of ideas, due to Freedman (1982), is that a Casson handle — no matter how complex the finite signed tree indexing its construction — is topologically indistinguishable from an open standard 2-handle.

Freedman (1982). *Let CH be any Casson handle. Then the interior of CH is homeomorphic to the interior of the standard 2-handle* $D^2 \times D^2$ *, that is, to* $D^2 \times \mathbb{R}^2$ *.*

$$\mathrm{CH} \underset{\text{homeo}}{\cong} D^2 \times \mathbb{R}^2$$

The proof rests on a sophisticated shrinking argument descended from Bing topology: one constructs, inside the Casson handle, a topological (but not smooth) flat disk that plays the role of the missing Whitney disk. As a topological space, the infinite nesting collapses to a plain $D^2 \times \mathbb{R}^2$.

This topological flattening underlies the classification of topological 4-manifolds — in particular, the complete classification of simply-connected topological 4-manifolds by their intersection form — and was recognized by the Fields Medal awarded to Freedman in 1986.

> **◇ Plain commentary: what 'topologically the same' means**
> Two spaces are homeomorphic if one can be continuously deformed into the other without cutting or gluing. The classical quip is that a coffee cup and a doughnut are homeomorphic — you can stretch one into the other.

> Freedman's theorem is a much deeper instance of the same idea: the Casson handle, despite its infinite tower of nested kinky handles, can be continuously deformed into a perfectly plain 4-dimensional cylinder $D^2 \times \mathbb{R}^2$. All of the visible complication — the infinite branching, the self-intersections — collapses under continuous deformation.
>
> This is what makes the Casson handle interesting. It is one of those mathematical objects whose 'shape' depends entirely on what category of equivalence one chooses. As a topological space, it is utterly plain. As a smooth manifold, it is anything but.

5.6 Donaldson's smooth obstruction

The smooth category sees what the topological category does not. Donaldson's theorem (1983) provides an obstruction to smoothness that has no topological analogue.

> **Donaldson (1983).** *Let X be a closed simply-connected smooth 4-manifold whose intersection form Q is negative definite. Then Q is diagonalizable over the integers.*

The proof studies the moduli space of solutions to the Yang–Mills anti-self-dual equations on an SU(2)-bundle over X. The geometry of this moduli space — in particular, the existence of a cobordism from a singular configuration to a manifold of reducible connections — translates into the diagonalizability claim. We will not retrace the argument here; for our purposes the consequence is what matters.

A direct consequence: the closed topological 4-manifold whose intersection form is the rank-16 form $E_8 \oplus E_8$ — even, negative definite, and not diagonalizable over $\mathbb{Z}$ — exists by Freedman but admits no smooth structure by Donaldson. The same logic applied to suitable open subsets produces non-standard smooth structures on $\mathbb{R}^4$.

> **◇ Why gauge theory enters**
>
> Donaldson's argument is striking precisely because it imports a tool from physics — the Yang–Mills equations of gauge theory — and uses it to derive a constraint on the smooth structure of a 4-manifold. In four dimensions the Yang–Mills moduli space has just the right dimension to produce information about the underlying smooth manifold; in other dimensions the analogous moduli spaces fail to have this property.

> The technical machinery is significant. Compactness of the moduli space requires the work of Uhlenbeck on gauge fixing; transversality requires perturbation arguments; the cobordism argument leading to diagonalizability requires careful analysis of singularities in the moduli space. None of this needs to be understood in detail for the present note.

> **◇ The intersection form and E8**
>
> The *intersection form* of a 4-manifold is a bilinear form that records, in homological terms, the intersection numbers of pairs of 2-dimensional submanifolds. It is the most basic algebraic invariant of a closed 4-manifold.
>
> **E8** is a negative-definite 8 × 8 symmetric matrix corresponding to the Dynkin diagram of the exceptional Lie algebra of the same name. It appears throughout Lie theory, lattice theory, and string theory.
>
> By Freedman's theorem there exists a unique closed simply-connected topological 4-manifold whose intersection form is $E_8 \oplus E_8$ — call it the *E8 manifold*. By Donaldson's theorem this manifold admits no smooth structure.
>
> Freedman therefore says 'this manifold exists topologically'; Donaldson says 'it does not exist smoothly'. The construction of exotic $\mathbb{R}^4$ extracts an object that lies in the gap between the two — an object that is topologically present but smoothly absent.

5.7 Construction of exotic ℝ⁴ and the uncountable family

The combination of Freedman and Donaldson produces exotic $\mathbb{R}^4$. The argument schema is as follows.

- Start with a closed simply-connected topological 4-manifold M whose intersection form makes it admit no smooth structure (by Donaldson), but which exists topologically (by Freedman). The simplest example involves Casson handles attached along a particular link.
- Freedman's theory yields a topological splitting M ≈ M1 ♮_top M2 along an embedded S^3.
- Try to realize this splitting smoothly. Along the way, the Donaldson obstruction prevents the smooth realization in certain regions.

- From the discrepancy between the topological splitting and the smooth obstruction, one extracts an open subset homeomorphic to $\mathbb{R}^4$ whose induced smooth structure is not diffeomorphic to the standard $\mathbb{R}^4$.

This first construction was made explicit by Casson and Freedman; subsequent work greatly enlarged the family. Gompf (1985) showed that end-connected sum produces an infinite family of mutually non-diffeomorphic exotic $\mathbb{R}^4$'s:

$$R_1 \natural R_2 \;=\; \big(R_1 \sqcup R_2\big)/\sim_{\text{ends}}$$

Taubes (1987) used a refined gauge-theoretic analysis on end-periodic 4-manifolds to demonstrate that the family in fact has cardinality of the continuum. DeMichelis–Freedman (1992) showed that this uncountable family can be realized as a family of open subsets of standard $\mathbb{R}^4$:

$$\iota : R_{\text{exotic}} \hookrightarrow \mathbb{R}^4_{\text{std}} \quad \text{(open embedding, not merely immersion)}$$

That is, inside the standard $\mathbb{R}^4$ there are uncountably many open subsets, each homeomorphic to $\mathbb{R}^4$ but each carrying a smooth structure pairwise non-diffeomorphic to the others and to the standard one.

5.8 Why this construction does not lift to finite computation

We can now state precisely what was claimed in §4.3.

The Casson handle is defined as a colimit of an increasing chain of kinky handles. There is no finite stage at which the construction terminates; the geometric object 'Casson handle' is essentially the entire infinite tower. Truncation at any finite stage gives a manifold whose interior is not yet homeomorphic to a standard 2-handle, so finite truncations are qualitatively different from the limit object.

The non-triviality of the Casson handle is drawn out by the tension between two heavyweight theories — Freedman's topological shrinking on one side, and Donaldson's analysis of the moduli of solutions to a nonlinear elliptic PDE on the other. Neither theory admits a presentation as a finite computational graph or a closed-form algebraic procedure.

The Freedman flattening theorem applies to the limit, not to any finite truncation. The

Donaldson obstruction is a global statement about a moduli space of PDE solutions on a closed manifold — there is no local or finite presentation of the obstruction that an autodiff framework could in principle compute.

Together: implementing exotic $\mathbb{R}^4$ in a contemporary numerical framework would require expressing both an infinite tower (the Casson handle) and a global PDE moduli space (the Donaldson obstruction) within the finite, local, symbolic vocabulary of computational graphs and IEEE 754 arithmetic. No part of this is impossible in principle, but no part of it is naturally expressible in any extant framework either. The standard smooth structure is, for the foreseeable computational landscape, the only smooth structure available.

> **◇ A summary for AI research scientists**
> Reduced to its essentials, the chapter has established the following:
>
> – The Casson handle is the geometric record of the failure of the Whitney trick in dimension four, organized as an infinite tower of kinky handles.
> – Freedman (1982) showed that this infinite tower is, after all, topologically standard — its interior is homeomorphic to a plain 4-dimensional cylinder.
> – Donaldson (1983) showed that smoothly the situation is different — there exist topological 4-manifolds that admit no smooth structure at all.
> – Combining the two yields exotic $\mathbb{R}^4$, and refined work by Taubes and DeMichelis–Freedman shows there are uncountably many of them, sitting inside the standard $\mathbb{R}^4$ as open subsets.
> – Both the Freedman flattening (a Bing-style infinite shrinking) and the Donaldson obstruction (a global PDE moduli analysis) are intrinsically infinite arguments, with no finite presentation suitable for current computational frameworks.
>
> We turn in §6 to a focused comparison of SGD and the natural gradient with respect to two distinct invariance properties, and in §7 to the broader significance of the invisibility result.

6. SGD versus the Natural Gradient: Two Notions of Invariance

Section 4 asserted, without elaboration, that the natural gradient method is no more sensitive to

the choice of smooth structure than ordinary SGD. This section makes the claim precise. The key step is to distinguish two notions of invariance that are sometimes conflated in the machine-learning literature: invariance under reparametrization, which the natural gradient does provide, and invariance under change of smooth structure, which it does not — and could not.

6.1 Reparametrization invariance

The principal advertised property of the natural gradient method is its invariance under reparametrization, in the sense established by Amari (1998). We state the property precisely.

Let Θ_1 and Θ_2 be two parameter spaces, and let $\psi : \Theta_1 \to \Theta_2$ be a smooth bijection. Suppose ψ is a reparametrization of the same statistical family — that is, $p(x; \theta_1) = p(x; \psi(\theta_1))$ for every $\theta_1 \in \Theta_1$. Then the Fisher metric transforms covariantly: $g_2 \circ d\psi = g_1$ on every tangent space, or equivalently

$$\varphi : (U, g_U) \xrightarrow{\sim} (V, g_V) \quad \Longrightarrow \quad \varphi_*\big(\mathrm{grad}_{g_U} L\big) \;=\; \mathrm{grad}_{g_V}(L \circ \varphi^{-1})$$

Consequently, the natural gradient flow on (Θ_1, g_1) is conjugate via ψ to the natural gradient flow on (Θ_2, g_2).

The most familiar manifestation of this invariance: a Gaussian family parametrized by mean μ and standard deviation σ produces, under the natural gradient, the same trajectories of distributions as the same Gaussian family parametrized by μ and variance σ^2. The change of variables $\sigma \mapsto \sigma^2$ is a smooth bijection that does not change the probability distributions involved, and the natural gradient is invariant under it.

> **◇ The practical relevance of reparametrization invariance**
> Vanilla SGD is sensitive to the choice of parametrization. Switching from σ to σ^2 rescales the gradient by a Jacobian factor, which changes the effective step size in that coordinate and alters the trajectory. The natural gradient cancels this Jacobian by construction, because the Fisher metric absorbs it.
>
> An important caveat: the 'reparametrizations' here are smooth bijections of the parameter space *that preserve the underlying statistical family*. Arbitrary smooth bijections of the parameter space need not preserve the family, and the natural gradient need not be invariant under them. The relevant invariance group is the diffeomorphism

> group of the statistical model, not of the parameter space as such.

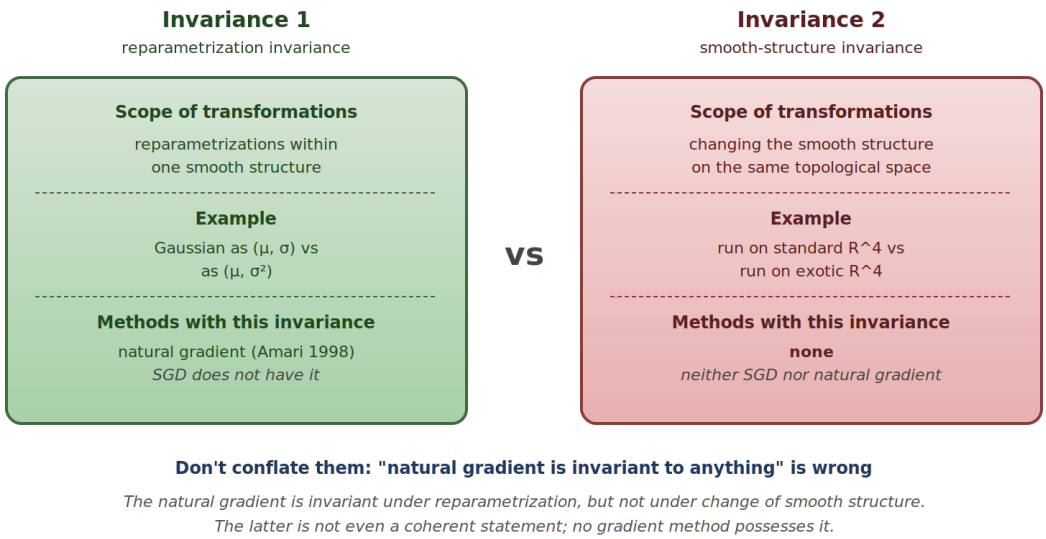
6.2 Smooth structure invariance does not exist

The reparametrization invariance of §6.1 is a statement about smooth bijections within a single smooth structure on the parameter space. There is a different question one could try to ask: given two parameter spaces that are homeomorphic but not diffeomorphic — say, standard $\mathbb{R}^4$ and an exotic $\mathbb{R}^4$ — does the natural gradient give the same trajectories on both?

The question is, on inspection, ill-posed. The natural gradient is defined as the steepest descent with respect to the Fisher metric, and the Fisher metric is a Riemannian metric on a smooth manifold. Comparing the Fisher metric on standard $\mathbb{R}^4$ with the Fisher metric on an exotic $\mathbb{R}^4$ requires a smooth bijection between them, and there is none — that is precisely what 'exotic' means. The natural gradient's invariance theorem cannot even be stated across smooth structures, let alone verified.

> **◇ Two notions of invariance, juxtaposed**
>
> **Invariance under reparametrization (Amari 1998).** A property of the natural gradient: smooth bijections of the parameter space that preserve the statistical family conjugate the natural gradient flow to itself. This is invariance *within* a single smooth structure on the parameter space.
>
> **Invariance under change of smooth structure.** Not a property of any gradient method, and not even a coherent statement: comparing two non-diffeomorphic smooth structures requires a smooth bijection between them, and by definition none exists. There is no algorithm — natural gradient or otherwise — that is invariant in this sense.
>
> The two notions are sometimes conflated in informal discussion. The natural gradient is sometimes described as 'invariant under arbitrary diffeomorphism', or as 'geometric and therefore basis-free'. These descriptions are correct but apply only within the smooth structure that has been fixed at the outset; they say nothing about what happens when the smooth structure itself is varied.

Figure 6. Two notions of "invariance", distinguished



6.3 Reconciliation with the framework of §§2-4

Putting §6 together with §§2-4, we have the following layered picture of what is and is not chosen by various components of a learning system.

- **Chosen by the implementation, before any algorithm is selected:** the smooth structure on the parameter space (the standard structure on $\mathbb{R}^N$, determined by IEEE 754 and autodiff).
- **Chosen by the algorithm, given the smooth structure:** the Riemannian metric (Euclidean for SGD/Adam, Fisher for the natural gradient).
- **Determined by the metric:** the gradient direction, the gradient flow, and the optimization trajectory.

The natural gradient improves on SGD at the second layer — it makes a more principled choice of metric — but does nothing at the first layer. The first layer's choice is not exposed by the algorithm, and the question (Q3) of §1.3 admits the same answer for both: neither SGD nor the natural gradient is sensitive to the choice of smooth structure on the parameter space.

7. Discussion: The Significance of Invisibility

We collect the main conclusions, address the question of what their significance is, and indicate

some speculative directions in which the framework might be extended.

7.1 Answering Q1, Q2, Q3

We restate the three questions of §1.3 with the answers reached over the course of the note.

- **(Q1)** Which smooth structure does the learning algorithm actually run on, and what determines that choice? — **Answer:** The standard smooth structure on $\mathbb{R}^N$, fixed by the IEEE 754 specification, automatic differentiation, and the wider computational stack. The optimization algorithm itself plays no role in this choice.
- **(Q2)** Can the choice of smooth structure change the optimum that training converges to? — **Answer:** In principle yes, in the dimension where alternative smooth structures exist. Two non-diffeomorphic smoothings of the same topological space have different algebras of smooth functions and thus different sets of critical points and different gradient flows. In practice no — the standard smooth structure is the only one available, so the difference is not realized in any current system.
- **(Q3)** Do different optimization algorithms differ in their sensitivity to the smooth structure? — **Answer:** No. SGD, Adam, and the natural gradient all sit downstream of the choice of smooth structure, and none provides a mechanism for varying that choice. The natural gradient's celebrated invariance is a statement about reparametrizations within a fixed smooth structure, not about changes of smooth structure.

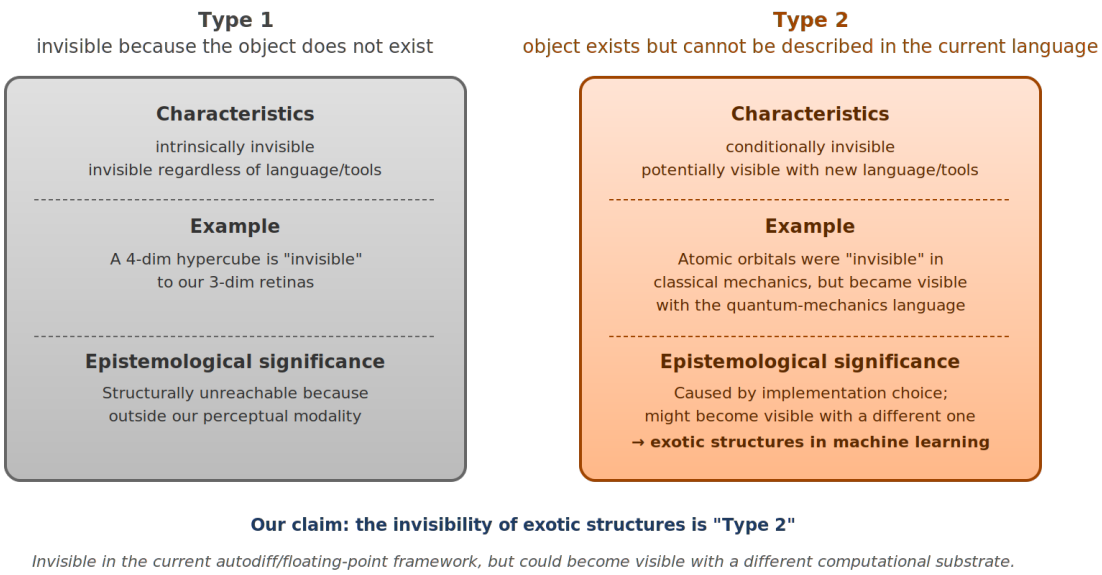
7.2 Two kinds of invisibility

The conclusion of the note is that exotic smooth structures are invisible to current machine learning. A natural reaction is that invisibility makes the issue not worth discussing. We argue against this reaction by distinguishing two kinds of invisibility.

- **Type 1: invisible because absent.** A four-dimensional hypercube is invisible to a three-dimensional eye not because of any defect in our optical equipment but because the object's full structure does not exist within our perceptual modality. The invisibility is intrinsic to the relationship between the object and the observer.
- **Type 2: invisible because not yet expressible.** Atomic electron orbitals were invisible to classical mechanics because the language of classical mechanics had no vocabulary for them. With the invention of quantum mechanics — a new language with a new vocabulary — the same orbitals became visible, computable, and predictable.

Exotic smooth structures are invisible to machine learning in the second sense, not the first. They exist mathematically; they are studied by a substantial community of differential topologists; their existence and properties are established with full rigor. What they lack is a representation in the vocabulary of contemporary numerical computation.

Figure 7. Two kinds of "invisibility"



This distinction matters because Type-2 invisibility is, by its nature, contingent on the choice of language. A change of computational paradigm — for instance, one that treats smooth structures as first-class objects rather than as implicit consequences of arithmetic primitives — could in principle make exotic structures visible. Should that happen, the questions of §1.3 would need to be re-examined: the answers we have given are answers within the contemporary paradigm, not necessary truths.

7.3 Tacit mathematical commitments

The deeper purpose of this note is to bring tacit mathematical commitments of machine learning into explicit view. The standard setup — 'parameter space $\mathbb{R}^N$, standard smooth structure, gradient descent' — is not the unique mathematically possible setup; it is a particular choice within a much larger space of possibilities. That standard setup happens to be supported by the computational substrate we have, and its competitors are not. This is a fact about computation, not about mathematics, and it is worth knowing as such.

Acknowledging tacit commitments has at least three concrete uses.

- It clarifies the scope of theorems. Convergence guarantees, generalization bounds, and information-geometric statements about the natural gradient are statements within the standard smooth structure. They are not statements about the parameter space as a topological object.
- It marks the boundary between mathematics and engineering. Casson handles, Donaldson invariants, and exotic $\mathbb{R}^4$ are mathematically real and well-studied objects that have no engineering presence in current systems. The boundary between the mathematically realizable and the engineering-realizable is itself a substantive object of study.
- It opens future questions. If a future computational framework supports alternative smooth structures, what kinds of exotic $\mathbb{R}^4$ are most useful to consider? Do trained models on different smooth structures generalize differently? Do they encode different representations? These are questions that cannot be asked today but that the present framework prepares one to ask.

7.4 Speculative directions

We conclude with three speculative directions of inquiry. None of them is well-developed, and all of them face substantial obstacles before they could be made operational.

7.4.1 Synthetic differential geometry

Synthetic differential geometry (Kock, Lawvere) provides a categorical formalism for smoothness in which infinitesimals are first-class objects. If automatic differentiation could be generalized to operate within such a framework, the smooth structure of the parameter space would become an explicit and selectable parameter rather than an implicit consequence of arithmetic. Whether this is computationally tractable is unclear; the existing literature on synthetic differential geometry is at the level of foundations rather than implementation.

7.4.2 A persistence-like invariant for smooth structure

Persistent homology infers topological invariants of a space from finite samples; one might ask whether some analogue could infer smooth-structure invariants from finite samples of a parameter space. No mature theory of this kind exists. The closest existing work is on topological invariants of loss landscapes (Horoi et al. 2022; Liu et al. 2024), which operates at the topological

rather than smooth level.

7.4.3 Physical analogies

Brans, Sloane, and Asselmeyer-Maluga have argued that exotic smoothness on a 4-dimensional spacetime can produce physical effects — additional gravitational sources, modifications to general relativity — that are not predicted by the standard smooth model. By analogy, one might speculate that gradient flows on an exotic $\mathbb{R}^4$, viewed from the standard structure, would carry 'spurious forces' not predicted by standard analysis. This is at present a metaphor rather than a quantitative prediction; we know of no implementation in which it could be tested.

8. Conclusion

We summarize the argument of the note in four points and a closing remark.

- **(C1)** The smooth structure on which a learning algorithm runs is not chosen by the algorithm. It is fixed by the implementation — by IEEE 754 floating-point arithmetic and automatic differentiation — and is the standard smooth structure on $\mathbb{R}^N$.
- **(C2)** In dimension four, $\mathbb{R}^4$ admits uncountably many distinct smooth structures (Freedman, Donaldson, Taubes, Gompf, DeMichelis–Freedman). This is a feature of dimension four alone; in every other dimension $\mathbb{R}^n$ admits a unique smooth structure.
- **(C3)** Modern machine learning does not see this multiplicity, not because the alternatives are uninteresting but because they cannot be expressed in the finite computational vocabulary of the implementation. The Casson-handle construction underlying exotic $\mathbb{R}^4$ has no finite presentation suitable for autodiff.
- **(C4)** This invisibility is itself a substantive observation. The standard smooth structure on $\mathbb{R}^N$ is not a self-evident default but a tacit choice embedded in the computational substrate, and acknowledging this clarifies what theorems about gradient methods do and do not say.

The note is, throughout, a thought experiment rather than a contribution to applied machine learning. Its purpose is to make visible a structural choice that is normally invisible, and to suggest that bringing such choices to the foreground is healthy for both the optimization-theoretic and the differential-geometric understanding of what gradient-based learning actually does. If the next time an AI research scientist writes $\theta \in \mathbb{R}^N$ on a whiteboard they pause, even briefly, to recall the uncountable family of smooth structures that the symbol silently selects against, the

note will have achieved its goal.

Appendix A: Corrections for Specialists in Differential Topology

The main text is pitched primarily at AI research scientists, and several passages sacrifice technical precision for accessibility. This appendix collects the technical corrections that a specialist in four-manifold topology or in the theory of exotic spheres would rightly want made. Each entry is in two parts: a precise statement of the correction, followed by a plain-language commentary that makes the correction accessible to AI research scientists.

A.1 The number 28 in Kervaire–Milnor

• Precise correction

In §3.2 we cited the result that S^7 admits 28 distinct smooth structures, referring to the theorem of Kervaire–Milnor (1963):

$$\Theta_7 \;\cong\; \mathbb{Z}/28\mathbb{Z}$$

Here Θ_n denotes the abelian group, under connected sum, of smooth structures on the oriented topological n-sphere up to orientation-preserving diffeomorphism. The number 28 thus counts *oriented* smooth structures. Forgetting orientation reduces the count — structures admitting an orientation-reversing self-diffeomorphism collapse to a single class under the forgetful map. More finely, there is a short exact sequence relating the subgroup bP_8 of structures bounding parallelizable manifolds to the quotient:

$$bP_8 \;\cong\; \mathbb{Z}/28\mathbb{Z}, \qquad \Theta_7/bP_8 \;\cong\; \pi_7^s/J$$

Here π_7^s is the 7th stable homotopy group of spheres and J is the J-homomorphism. The main-text statement is not wrong in its number, but it is imprecise about what the number is counting.

> **◇ Plain-language version**
>
> The number 28 is correct, but exactly what it counts takes a moment to state. The picture is this.
>
> One can 'add' two smooth structures on S^7 by a construction called connected sum — essentially, cut a small disk out of each, glue the two boundary spheres together, and keep

> the result. This addition satisfies the axioms of a group.
>
> That group is Θ_7, and Kervaire and Milnor proved it is cyclic of order 28 — the integers modulo 28.
>
> But whether two smooth structures count as 'the same' depends on whether we keep track of orientation. If the structures are considered up to orientation-preserving diffeomorphism, there are 28. If we allow orientation-reversing diffeomorphisms to identify structures, the count is smaller.
>
> So the literal statement 'S^7 has 28 smooth structures' is only exactly right in the oriented setting.
>
> For ML readers: it is a little like whether two trained models count as 'the same' when one allows a symmetry (permuting neurons) versus requiring exact identity. The count of equivalence classes depends on which equivalence relation one imposes.

A.2 A more precise definition of a kinky handle

- **Precise correction**

The schematic we used in §5.3,

$$K \;\cong\; (D^2 \times D^2) \natural \big(\text{self-plumbings}\big)$$

is an abbreviation rather than a definition. Rigorously, a kinky handle K is obtained by taking the standard 2-handle $D^2 \times D^2$ and replacing its embedded core by an immersion whose image has exactly d transverse double points (all of the same sign, by convention).

Boundary connected summing with collars at each double point produces a 4-manifold that, up to homotopy equivalence, is a 2-handle with d one-handles attached along its boundary. Formally,

$$K \;=\; (D^2 \times D^2) \;\cup_\partial\; \big(\text{1-handles at double points}\big)$$

Additionally, the main text's passing reference to 1-handles of type $\partial D^2 \times D^2$ is a misstatement; the correct type of 1-handle attached for each self-plumbing is $S^1 \times D^3$ (the main text is correct on this point elsewhere).

> **◇ Plain-language version**
> The main text described a kinky handle as 'a 2-handle whose core has been allowed to self-intersect'. That is correct in spirit, but the formal definition has more structure.
>
> Imagine a clean 2-disk inside a 4-dimensional region. The kinky handle is what you get when you replace that clean disk by a slightly twisted one whose interior crosses itself at exactly d points (the kink number).
>
> For each crossing, there is a small adjustment — a 'collar' added to the boundary — and the result is, up to homotopy, the same as a 2-handle with d separate handles attached at its rim.
>
> These distinctions matter for proofs but not for the qualitative picture.

A.3 The branching structure of a Casson tower

• Precise correction

In §5.4 we referred to the tree indexing a Casson tower as a 'signed binary tree'. This is incorrect.

The correct term is 'finitely-branching signed tree': the number of branches leaving a given vertex equals the self-plumbing count d of the kinky handle at that vertex, and d is an arbitrary non-negative integer, not restricted to 2. The signs decorate the double points.

A more formal inductive statement of the construction is

$$\mathrm{CH} \;=\; \bigcup_{k\geq 0} K_k, \qquad K_k \subset K_{k+1}, \quad \partial^- K_{k+1} = \partial^+ K_k$$

where K_{k+1} consists of finitely many kinky handles attached to K_k. The combinatorial type of the tree (the branching factor d_v at each vertex v and the sign at each double point) determines the isometry class of the Casson handle.

> **◇ Plain-language version**
> The Casson tower's branching is not strictly two at each level. It is finitely many, with the number depending on the kink count of the handle at that level.

Concretely:

- Each kinky handle has some number d ≥ 0 of self-intersections.
- At the next level of the tower, d new kinky handles are attached, one at each self-intersection.
- Each of those new handles has its own kink count, which determines the branching at the level after that.

The result is a finitely-branching tree, not a binary tree. The signs (+ or –) attached to the edges record the orientation of the corresponding self-intersections.

An analogy for ML readers: just as a decision tree is not always a binary tree, the branching of a Casson tower is in general arbitrary, not binary.

A.4 The precise statement of Freedman's theorem

• Precise correction

In §5.5 we asserted that 'the Casson handle is topologically the same as an ordinary 2-handle.' The strictly correct statement is that the *interior* of the Casson handle is homeomorphic to the interior of the standard 2-handle, that is,

$$\mathrm{int}(\mathrm{CH}) \underset{\text{homeo}}{\cong} \mathrm{int}(D^2 \times D^2) \cong \mathbb{R}^2 \times \mathbb{R}^2$$

(Freedman 1982, Theorem 1.1). The boundary behavior is more delicate. The proof uses Bing-style topological shrinking, applied to a decomposition of the Casson handle that produces a topological flat disk inside it; this flat disk is what plays the role of the missing Whitney disk.

◇ Plain-language version

The main text says 'a Casson handle is topologically the same as an ordinary 2-handle.' The strictly correct version says 'the interior of a Casson handle is the same as the interior of an ordinary 2-handle.' The distinction is real and familiar:

- The open interval (0, 1) and the closed interval [0, 1] have the same interior but different boundary behavior.
- In the same way, Casson handles and the standard 2-handle agree in their interior but differ at the boundary in subtler ways.

> Exotic $\mathbb{R}^4$ is an open manifold, and the interior statement is all that this construction needs. The main text's looser wording does not affect the story for ML.

A.5 The simply-connected hypothesis in Donaldson's theorem

• Precise correction

Our §3.3.2 / §5.6 formulation of Donaldson's theorem omitted a hypothesis. The 1983 diagonalization theorem applies to simply-connected manifolds:

$$X \text{ simply connected, closed, oriented, smooth 4-manifold with } Q_X \text{ negative definite} \implies Q_X \cong \bigoplus \langle -1 \rangle$$

Dropping simple connectivity gives a different problem whose resolution (for various finite and infinite fundamental groups) unfolded over subsequent decades; Furuta's 10/8 theorem and the work of Fintushel and Stern are representative of the post-1983 progress. For our application — the construction of exotic $\mathbb{R}^4$ — the relevant 4-manifolds (the K3 surface and its relatives) are simply connected, so the omission is harmless in context. It is nonetheless an omission.

> **◇ Plain-language version**
>
> Donaldson's 1983 theorem carries a hypothesis we did not state in the main text: the manifold must be simply connected. 'Simply connected' means every loop in the space can be continuously shrunk to a point. Examples:
>
> – S^4 is simply connected: any loop on the sphere can be pulled tight to a single point.
> – A torus (the surface of a doughnut) is not simply connected: a loop going once around the hole cannot be pulled off.

> Donaldson's original result applies only in the simply-connected setting. The manifolds relevant to the construction of exotic $\mathbb{R}^4$ — K3 and its relatives — are simply connected, so in our context the hypothesis is free. Extending Donaldson-type obstructions to non-simply-connected 4-manifolds required further decades of work (Furuta, Fintushel–Stern, and others).

A.6 The construction of exotic $\mathbb{R}^4$ stated precisely

• Precise correction

In §5.7 we wrote that 'Freedman's theory yields a topological splitting M ≈ M_1 ♮_top M_2' of a certain closed 4-manifold M. This is an oversimplification. The canonical Casson–Freedman construction is more naturally stated in the language of h-cobordism:

- (i) Freedman's topological h-cobordism theorem shows that in the topological category, 5-dimensional h-cobordisms between simply-connected 4-manifolds are trivial.
- (ii) Donaldson (1987) exhibited a smooth 5-dimensional h-cobordism that is not a smooth product, proving that the smooth h-cobordism theorem fails in dimension four.
- (iii) The discrepancy between what is topologically trivial and what is smoothly non-trivial manifests, on suitable open subsets, as exotic $\mathbb{R}^4$.

Our choice of example — 'a blowup of a K3 surface' — is also loose. The classical literature tends to use

$$\mathbb{CP}^2 \# 9\,\overline{\mathbb{CP}^2}, \quad E(2) = \mathrm{K3}, \quad E(n)$$

or elliptic surfaces E(n); the K3 surface itself is E(2), so speaking of 'a blowup of K3' can mislead. Finally, Taubes (1987) proved that the existing family of exotic $\mathbb{R}^4$ is mutually non-diffeomorphic using end-periodic ASD moduli; the uncountability itself was obtained by combining this with Gompf's (1985) end-sum construction. The main text's prose blurred the precedence of these contributions.

> **◇ Plain-language version**
>
> The main text's account of the exotic-$\mathbb{R}^4$ construction is correct in outline but imprecise in three respects:
>
> – 'Connected-sum splitting' is too crude a description. The right setting is 'h-cobordism', a concept that captures when two manifolds are connected by a 'collar' — in higher dimensions the collar can always be straightened, but in dimension four it cannot.
>
> – 'Blowup of a K3 surface' is not the standard construction. The classical examples are $E_8 \oplus E_8$ or the elliptic surface E(n); the K3 surface is itself E(2), and there is no 'blowup' step needed.
>
> – The uncountability of exotic $\mathbb{R}^4$'s comes from two distinct results in tandem: Gompf (1985) generated the family by end-sum, and Taubes (1987) showed that the members of the family are pairwise non-diffeomorphic. The main text did not separate these two contributions.

> For ML readers: these are internal matters of mathematics. The conclusion — 'exotic $\mathbb{R}^4$ is uncountable in number' — is unchanged. The implications for ML are not affected.

A.7 Relation to the smooth 4-dimensional Poincaré conjecture

● Precise correction

We made a brief reference in §3.1 to 'the question of exotic structures on S^4 is open'. To make the statement precise: the smooth 4-dimensional Poincaré conjecture and the question of exotic S^4 are not strictly equivalent, but they are related by an inclusion.

- The smooth 4-dimensional Poincaré conjecture states: every smooth 4-manifold homotopy equivalent to S^4 is diffeomorphic to S^4.
- The exotic-S^4 question asks whether S^4 admits any smooth structure not diffeomorphic to the standard one.

In four dimensions the topological Poincaré conjecture (proved by Freedman 1982) implies that any topological 4-manifold homotopy equivalent to S^4 is homeomorphic to S^4. Combined with the topological-vs-smooth distinction, this makes the smooth Poincaré conjecture equivalent to the non-existence of exotic S^4.

> **◇ Plain-language version**
>
> We mentioned in passing that 'the question of exotic S^4 is open'. Strictly speaking this is correct, but the relationship to the smooth Poincaré conjecture deserves to be spelled out.
>
> Step 1: 'Is every topological 4-manifold homotopy equivalent to S^4 actually homeomorphic to S^4?' — this is the topological 4-dimensional Poincaré conjecture. Freedman resolved it in 1982, with the answer Yes.
>
> Step 2: 'Is every smooth 4-manifold homeomorphic to S^4 also diffeomorphic to S^4?' — this is the smooth 4-dimensional Poincaré conjecture. It is unresolved.
>
> Because the first step is settled, the second becomes equivalent to the question of whether S^4 admits any exotic smooth structure. The main text combined the two into a single phrase; a careful treatment states them separately.

> In higher dimensions the situation differs. Smooth Poincaré conjectures are resolved in many dimensions, sometimes positively and sometimes negatively — S^7 admits exotic structures, while S^4 5, S^4 6, and S^4 12 do not. Dimension four alone remains open.

A.8 The DeMichelis–Freedman result, stated precisely

• Precise correction

In the main text we wrote that 'the family of exotic $\mathbb{R}^4$'s living in standard $\mathbb{R}^4$ is uncountable.' The accurate statement uses *embedding* (not *immersion*). The DeMichelis–Freedman (1992) theorem asserts:

$$\iota : R_{\text{exotic}} \hookrightarrow \mathbb{R}^4_{\text{std}} \quad \text{(open embedding, not merely immersion)}$$

that is, an uncountable family of open subsets of standard $\mathbb{R}^4$, each smoothly embedded (no self-intersection), each homeomorphic to $\mathbb{R}^4$, and pairwise non-diffeomorphic in the smooth structure inherited from the embedding. 'Immersion' would allow self-intersections and is the wrong notion here.

> **◇ Plain-language version**
> The distinction between 'embedding' and 'immersion' is the following:
>
> – An *embedding* places a smaller space inside a larger one without self-intersection — for instance, knotting a circle inside 3-dimensional space.
> – An *immersion* places a smaller space inside a larger one allowing self-intersections — for instance, immersing a Klein bottle in 3-dimensional space.
>
> DeMichelis–Freedman (1992) showed that exotic $\mathbb{R}^4$'s sit inside standard $\mathbb{R}^4$ as honest open subsets — embeddings, no self-intersections — and the embedded copies are pairwise non-diffeomorphic. 'Embedding' is the strictly stronger and correct word.

Appendix B: Corrections for Specialists in Information Geometry

The following appendix lists the technical corrections that an information-geometer reading the main text would want addressed. As in Appendix A, each entry has a precise statement followed by a plain-language commentary.

B.1 The precise scope of natural-gradient invariance

● Precise correction

In §6.1 we stated the invariance of the natural gradient under a smooth bijection ψ : Θ_1 → Θ_2 . This is imprecise. The correct statement requires that ψ be not merely a diffeomorphism but a reparametrization of the same statistical model — that is, p(x; θ _1) = p(x; ψ(θ _1)) for every θ _1 ∈ Θ_1. Then the Fisher metric pushes forward correctly:

$$\psi : (\Theta_1, g_1^F) \xrightarrow{\sim} (\Theta_2, g_2^F), \qquad p(x;\theta_1) = p(x;\psi(\theta_1))$$

and the natural gradient flow on (Θ_1, g_1) is conjugate via ψ to that on (Θ_2, g_2). Under arbitrary smooth bijections of the parameter space that do not preserve the statistical family, this conjugacy fails. The main text was loose on this point.

As a supplement, when the statistical model is regular, the Fisher information matrix admits the Hessian representation

$$g_{ij}^F(\theta) \;=\; -\,\mathbb{E}_{x\sim p(x;\theta)}\left[\frac{\partial^2 \log p(x;\theta)}{\partial\theta^i\,\partial\theta^j}\right]$$

which directly reflects this statistical equivalence: if the distribution is unchanged under reparametrization, so is the right-hand side.

> **◇ Plain-language version**
>
> Two notions of 'reparametrization' are easily conflated:
>
> – Loose version: changing parameters leaves the answer unchanged.
> – Strict version: changing parameters in a way that does not change the probability distribution leaves the answer unchanged.
>
> Concrete example: a Gaussian can be parametrized by mean μ and standard deviation σ, or by mean μ and variance σ^2. Switching between these parametrizations does not change the underlying distribution, so the natural gradient is invariant under the switch.
>
> Arbitrary smooth coordinate changes, however, can easily change the underlying distribution; under those, natural-gradient invariance fails. The main text's phrase 'any diffeomorphism' is broader than what actually holds.

> For ML readers: when the literature speaks of 'reparametrization invariance', the precise meaning is invariance under reparametrizations that preserve the probability distribution. Worth keeping in mind.

B.2 Regularity hypotheses for the Fisher metric

• Precise correction

The Fisher information we wrote in §1.2,

$$G(\theta) \;=\; \mathbb{E}_{x\sim p(x;\theta)}\big[\nabla_\theta \log p(x;\theta)\,\nabla_\theta \log p(x;\theta)^\top\big]$$

(the score covariance) agrees with the Hessian form in B.1 only under regularity of the parametric model p(x; θ) — specifically, when differentiation and integration may be interchanged in the log-likelihood. For deep neural network conditional distributions p(y|x; θ) with piecewise-linear activations (ReLU, leaky ReLU, and so on), regularity can fail on measure-zero subsets of θ; in practice this almost never affects a training trajectory. More seriously, deep models are known to be *singular* statistical models in the sense of Watanabe (2009) — the Fisher information matrix degenerates (rank drops) along subvarieties of parameter space. The Fisher-metric framework used in classical information geometry excludes such singularities from its scope. Our discussion implicitly assumes regularity throughout.

> **◇ Plain-language version**
>
> The Fisher information matrix has two standard forms:
>
> – First-order form: the covariance of the gradient of the log-likelihood (the score function).
> – Second-order form: minus the expected Hessian of the log-likelihood.
>
> The two forms agree only for regular models — roughly, models for which differentiation passes under the integral. Gaussians are regular; deep networks with ReLU activations are not quite, because the activation fails to be differentiable at zero. In practice this does not affect training.
>
> A deeper issue lies in Watanabe's theory of singular statistical models, which draws on algebraic geometry and classifies deep networks as singular. At certain parameter settings the Fisher information matrix drops rank — a degenerate direction — and the classical Fisher-metric picture fails there.

> The present note sets this aside for simplicity. Exotic smoothness and statistical singularity are separate issues, though a unified treatment would be mathematically appealing.

B.3 The smooth structure underlying information geometry

● Precise correction

In §6.3 we asserted that the parameter space Θ in information geometry is 'an open subset of Euclidean space, with its standard smooth structure'. This accurately describes the working setting.

More abstract formulations (Chentsov's classification theorem, Amari–Nagaoka's α-connection theory) treat the statistical model itself as an abstract manifold and equip it with a smooth structure intrinsically.

Both the concrete and the abstract formulations silently fix a diffeomorphism class once and for all; neither contemplates the possibility of exotic smoothness. Our conclusion — that information geometry renders exotic structure invisible — therefore holds in both formulations.

Chentsov's theorem (Chentsov 1982) is worth flagging explicitly: up to a positive scalar, the Fisher metric is the unique Riemannian metric on a statistical model invariant under sufficient statistics. This is a uniqueness result for the *metric*, not for the underlying smooth manifold; it presupposes a smooth structure rather than constraining one.

> **◇ Plain-language version**
>
> Information geometry comes in two styles:
>
> – Concrete: treat the parameter space Θ as an open subset of Euclidean space with the standard smooth structure.
> – Abstract: treat the statistical model itself as an abstract manifold and put a smooth structure on it directly.
>
> In both styles, the standard practice fixes one smooth structure and works within it. The note's claim — that information geometry is insensitive to smooth structure — therefore applies to both styles equally.

> Chentsov's theorem, sometimes cited as a strong uniqueness result, states that the Fisher metric is the unique Riemannian metric on a statistical model respecting sufficient statistics. The uniqueness is of the metric, not of the underlying smooth manifold; both theorems presuppose a smooth structure before they can even be stated. Chentsov thus does not obstruct the existence of alternative smooth structures — it simply takes one for granted.
>
> The short version for ML readers: by construction, information geometry is not an instrument for detecting differences in smooth structure.

B.4 The natural gradient and smooth structure: logical order

• Precise correction

Our §6.2 statement — 'the natural gradient does not detect the choice of smooth structure' — is correct as a conclusion but compresses its logic.

The honest statement is this: the natural-gradient procedure is specified in three stages. (i) Fix a smooth structure on the parameter space. (ii) Define the Fisher metric with respect to that structure. (iii) Compute the steepest-descent direction with respect to that metric.

The choice of smooth structure enters only at stage (i); stages (ii) and (iii) presuppose (i) and leave it untouched. The natural gradient is thus insensitive to smooth structure not because its formalism detects and ignores differences, but because its formalism has nothing at which to look — the choice precedes the formalism.

> **◇ Plain-language version**
>
> The main text says 'the natural gradient does not detect the difference between smooth structures'. Correct as a conclusion, but the logical structure is worth spelling out. Applying the natural gradient requires three steps, in order:
>
> – Fix a smooth structure on the parameter space — typically the standard one on $\mathbb{R}^n$.
> – Define the Fisher metric with respect to that smooth structure.
> – Compute the steepest-descent direction relative to that metric.
>
> The smooth-structure question is settled at the first step. The later steps take that choice as given and cannot revisit it.

> So 'the natural gradient is blind to smooth structure' is better phrased as: the natural gradient is defined downstream of the choice of smooth structure, and the choice is simply not an object in its formalism.
>
> The practical takeaway is unchanged: natural-gradient training gives no new purchase on exotic $\mathbb{R}^4$.

Summary of the appendices

None of the corrections above alters the note's main conclusion — that exotic smooth structures are invisible within the current apparatus of machine learning.

They are offered for specialist readers who would otherwise be uncomfortable with the main text's simplifications. The author welcomes further corrections and refinements; please file them as GitHub issues.

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The author welcomes comments, corrections, and extensions, which may be submitted as GitHub issues.